# Railway Safety Intelligence System

## End-to-End Data Analytics & Machine Learning Project Report

Tools Used: Python, Pandas, NumPy, SQL, Power BI, Machine Learning, SHAP Explainable AI

**Datasets:**

**FRA Railroad Accident & Incident Data (USA)**

[Railroad Accident and Incident Data Dataset]

**Indian Railways Accidents (1902–2024)**

[Indian Railways Accidents Dataset]

## 1. Project Overview

Railway transportation is one of the most critical modes of transportation worldwide. Railway accidents can lead to significant financial losses, injuries, fatalities, operational disruptions, and infrastructure damage.

This project develops a Railway Safety Intelligence System that analyzes accident patterns, identifies risk factors, predicts accident severity, and provides interactive dashboards for decision-makers.

**The project combines:**

- ⁎ Data Cleaning
- ⁎ Exploratory Data Analysis (EDA)
- ⁎ SQL Analysis
- ⁎ Machine Learning
- ⁎ Explainable AI (SHAP)
- ⁎ Power BI Dashboards

using both U.S. FRA railway accident data and Indian Railway accident data.

## 2. Business Problem

**Railway authorities face challenges such as:**

- ⁎ Increasing accident rates
- ⁎ High maintenance costs
- ⁎ Human operational errors
- ⁎ Track-related failures
- ⁎ Delayed rescue operations
- ⁎ Difficulty identifying accident-prone regions

**The objective is to build a data-driven system that helps authorities:**

- ⁎ Monitor accident trends
- ⁎ Identify high-risk locations
- ⁎ Analyze causes of accidents
- ⁎ Predict accident severity
- ⁎ Improve safety planning

## 3. Project Objectives

### FRA Dataset

- ⋇ Analyze accident trends over time
- ⋇ Identify accident hotspots
- ⋇ Determine major accident causes
- ⋇ Study fatalities and injuries
- ⋇ Build accident severity prediction model

### Indian Railway Dataset

- ⋇ Analyze historical accident patterns
- ⋇ Compare accident frequency across states
- ⋇ Study rescue time effectiveness
- ⋇ Analyze railway safety fund utilization
- ⋇ Generate strategic safety insights

## 4. Dataset Description

### Dataset 1: FRA Railroad Accident Data

**Original Size**

218,361 Records
160 Columns

**After Cleaning**

217,619 Records
57 Columns

### Key Variables

| | |
|---|---|
| Accident Number | Latitude |
| Date | Longitude |
| State Name | Train Speed |
| County Name | Damage Cost |
| Accident Type | Fatalities |
| Accident Cause | Injuries |

### Dataset 2: Indian Railway Accidents

**Accidents Sheet:** 290 Historical Accident Records
**Funds Sheet:** 17 Years of Safety Fund Data
**Rescue Time Sheet:** Rescue Performance Metrics

## 5. Data Dictionary Creation

Before performing data cleaning and analysis, a comprehensive data dictionary was created to understand the structure, meaning, and business relevance of each variable across both datasets.

### Purpose

**The data dictionary was developed to:**

- ⋇ Understand column definitions and business context
- ⋇ Identify relevant variables for analysis
- ⋇ Classify columns into categories
- ⋇ Detect redundant and low-value attributes
- ⋇ Support data cleaning and feature engineering
- ⋇ Improve consistency across Python, SQL, and Power BI workflows

## FRA Dataset Data Dictionary

The original FRA dataset contained 160 columns.

**Columns were categorized into the following groups:**

### Accident Information

- ⋇ Accident Number
- ⋇ Date
- ⋇ Accident Year
- ⋇ Accident Month
- ⋇ Accident Type
- ⋇ Accident Cause
- ⋇ Narrative

### Geographic Information

- ⋇ State Name
- ⋇ Country Name
- ⋇ Division
- ⋇ Station
- ⋇ District
- ⋇ Latitude
- ⋇ Longitude

### Train Operations

- ⋇ Train Speed
- ⋇ Maximum Speed
- ⋇ Gross Tonnage
- ⋇ Equipment Type
- ⋇ Train Direction

### Environmental Factors

- ⋇ Temperature
- ⋇ Visibility
- ⋇ Weather Condition

### Safety Impact Metrics

- ⋇ Total Persons Killed
- ⋇ Total Persons Injured
- ⋇ Persons Evacuated

### Human Factors

- ⋇ Hours Engineers On Duty
- ⋇ Hours Conductors On Duty
- ⋇ Positive Alcohol Tests
- ⋇ Positive Drug Tests

### Rolling Stock Information

- ⋇ Loaded Freight Cars
- ⋇ Empty Freight Cars
- ⋇ Derailed Freight Cars
- ⋇ Locomotive Counts

### Financial Impact Metrics

- ⋇ Equipment Damage Cost
- ⋇ Track Damage Cost
- ⋇ Total Damage Cost

# Indian Railway Dataset Data Dictionary

## Accidents Sheet

Variables were grouped into:

**Accident Details**

- ⋇ Accident Name
- ⋇ Train Name
- ⋇ Accident Type
- ⋇ Standard Accident Type

**Location Information**

- ⋇ State
- ⋇ Location
- ⋇ Railway Zone

**Impact Metrics**

- ⋇ Deaths
- ⋇ Injuries
- ⋇ Rescue Time (hrs)

**Cause Analysis**

- ⋇ Cause
- ⋇ Standard Cause Type

**Corrective Measures**

- ⋇ Reforms/Changes

## Funds Allocated Sheet

**Funding Metrics**

- ⋇ Railway Safety Fund Total Grant
- ⋇ Railway Safety Fund Actual Expenditure
- ⋇ Depreciation Reserve Fund
- ⋇ Actual Utilisation

**Operational Expenditure Metrics**

- ⋇ Total Working Expenditure
- ⋇ Ordinary Working Expense

## Rescue Time Sheet

**Emergency Response Metrics**

- ⋇ Rescue ID
- ⋇ Accident Severity
- ⋇ Location
- ⋇ Average Rescue Time (hrs)

## Outcome of Data Dictionary Creation

**The data dictionary helped in:**

- ⋇ Selecting 57 high-value columns from the FRA dataset
- ⋇ Identifying irrelevant and highly sparse attributes
- ⋇ Standardizing column names
- ⋇ Designing SQL queries
- ⋇ Creating Power BI data models
- ⋇ Supporting feature engineering and machine learning development

This step significantly improved data quality, analytical efficiency, and project maintainability throughout the Railway Safety Intelligence System development lifecycle.

# 6. Data Cleaning Process

## FRA Dataset

### Cleaning Activities

- ⋇ Removed duplicate records
- ⋇ Standardized column names
- ⋇ Converted dates into datetime format
- ⋇ Converted numeric columns
- ⋇ Treated missing values
- ⋇ Removed impossible latitude and longitude values
- ⋇ Optimized memory usage
- ⋇ Generated quality reports

### Result

**Memory reduced from:** 1194 MB
**To:** 704 MB
**Approximate reduction:** 41%

## Indian Railway Dataset

### Cleaning Activities

- ⋇ Removed duplicate entries
- ⋇ Standardized state and zone names
- ⋇ Fixed data types
- ⋇ Treated missing values
- ⋇ Created year-based features
- ⋇ Cleaned safety fund data
- ⋇ Cleaned rescue time information

## 7. SQL Analysis

More than 25 SQL queries were executed.
Examples:

### Accident Trends

```
SELECT accident_year, COUNT(*) AS total_accidents FROM railway_safety
GROUP BY accident_year;
```

### Top States

```
SELECT state_name, COUNT(*) AS accidents FROM railway_safety GROUP BY
state_name ORDER BY accidents DESC;
```

### Damage Cost Analysis

```
SELECT accident_cause, SUM(total_damage_cost) FROM railway_safety GROUP
BY accident_cause;
```

Key findings were later integrated into Power BI dashboards.

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## 8. Exploratory Data Analysis (EDA)

### FRA Dataset Findings

#### Accident Trends

- Highest accident frequency occurred during earlier decades.
- Significant reduction observed after safety regulations improved.

#### Top Accident States

- Illinois
- Texas
- California
- Pennsylvania

#### Common Accident Types

- Derailments
- Side Collisions
- Highway-Rail Crossing Incidents

#### Major Causes

- Track Defects
- Equipment Failure
- Human Error

### Indian Dataset Findings

#### Accident Distribution

**Highest accident occurrence observed in:**

- Uttar Pradesh
- Andhra Pradesh
- Assam
- Maharashtra

### Railway Zones

**High accident concentration observed in:**

- ※ Northern Railway
- ※ Southern Railway
- ※ East Central Railway

### Safety Funds

Fund allocation increased consistently after 2015.

### Rescue Operations

Average rescue time: Approximately 5 hours

## 9. Feature Engineering

Additional features created:

### FRA Dataset

- ※ Accident Year Clean
- ※ Accident Month Clean
- ※ Accident Weekday

### Indian Dataset

- ※ Accident Decade
- ※ Fund Utilization Percentage
- ※ Rescue Efficiency Score

These features improved analytical and predictive capabilities.

## 9. Machine Learning Model

### Objective

Predict accident severity.

### Target Variables

- ※ Fatalities
- ※ Injuries
- ※ Damage Cost

### Algorithms Tested

- ※ Random Forest
- ※ Logistic Regression
- ※ XGBoost

### Model Workflow

1. Feature Selection
2. Train-Test Split
3. Model Training
4. Performance Evaluation
5. Feature Importance Analysis

## 11. Explainable AI (SHAP)

SHAP was used to explain model predictions.

### Benefits

- ⋇ Transparent predictions
- ⋇ Feature impact analysis
- ⋇ Regulatory compliance
- ⋇ Better stakeholder trust

### Important Features Identified

- ⋇ Train Speed
- ⋇ Track Density
- ⋇ Accident Cause
- ⋇ Weather Conditions
- ⋇ Visibility
- ⋇ Equipment Damage Cost etc.

## 12. Geographic Hotspot Detection

Geospatial analysis identified:

### FRA Hotspots

- ⋇ Illinois
- ⋇ Texas
- ⋇ California

### Indian Hotspots

- ⋇ Uttar Pradesh
- ⋇ Bihar
- ⋇ Assam

These regions require additional safety monitoring.

## 13. Power BI Dashboard

### Dashboard 1:

### Executive Safety Overview (FRA)

**KPIs:**

- ⋇ Total Accidents
- ⋇ Total Deaths
- ⋇ Total Injuries
- ⋇ Total Damage Cost

**Visuals:**

- ⋇ Accident Trend Analysis
- ⋇ Top States
- ⋇ Accident Type Distribution
- ⋇ Damage Cost by Cause

### Dashboard 2:

### Indian Railway Safety Insights

**KPIs:**

- ⋇ Total Accidents
- ⋇ Total Deaths
- ⋇ Total Injuries
- ⋇ Average Rescue Time

**Visuals:**

- ⋇ Accidents by State
- ⋇ Accidents by Railway Zone
- ⋇ Safety Fund Allocation
- ⋇ Rescue Time Analysis

## 14. Key Business Insights

### FRA Dataset

- ⁎ Derailments represent the largest category of accidents.
- ⁎ Track defects contribute significantly to incidents.
- ⁎ Certain states consistently experience higher accident volumes.
- ⁎ Accident frequency has reduced over time due to safety improvements.

### Indian Dataset

- ⁎ Accident concentration exists in specific states.
- ⁎ Safety fund allocation has increased significantly.
- ⁎ Rescue operations generally occur within a few hours.
- ⁎ Infrastructure modernization has improved safety outcomes.

## 15. Recommendations

### Infrastructure

- ⁎ Increase track inspections.
- ⁎ Upgrade aging railway infrastructure.
- ⁎ Deploy predictive maintenance systems.

### Operations

- ⁎ Strengthen employee safety training.
- ⁎ Improve fatigue monitoring.
- ⁎ Increase signal monitoring.

### Technology

- ⁎ Deploy AI-based accident prediction systems.
- ⁎ Use real-time monitoring sensors.
- ⁎ Expand geographic risk monitoring.

### Rescue Management

- ⁎ Reduce rescue response time.
- ⁎ Improve emergency coordination.
- ⁎ Establish regional rapid-response teams.

## 16. Project Outcome

**Successfully developed a complete Railway Safety Intelligence System that:**

- ⁜ Cleaned and processed large-scale railway datasets.
- ⁜ Conducted advanced EDA.
- ⁜ Performed SQL-based analytical reporting.
- ⁜ Built machine learning prediction models.
- ⁜ Applied SHAP Explainable AI.
- ⁜ Identified accident hotspots.
- ⁜ Developed interactive Power BI dashboards.
- ⁜ Generated actionable safety recommendations.

## 17. Technologies Used

- ⁜ Python
- ⁜ Pandas
- ⁜ NumPy
- ⁜ SQL
- ⁜ Power BI
- ⁜ Scikit-Learn
- ⁜ SHAP
- ⁜ Excel

## 18. Conclusion

This project demonstrates how data analytics, machine learning, and business intelligence can be combined to improve railway safety, reduce accident risks, optimize resource allocation, and support data-driven decision-making for railway authorities.